Physical Regularity Implies Inference-Like Dynamics in Graph Rewrite Systems

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Abstract.

We enumerate all 16 connected DPO graph rewrite rules at signatures $\sigma = 1$ through $\sigma = 4$ and classify each by two independent geometric predicates: the I-predicate (convergent state-space trajectories with compressible dynamics) and the Φ-predicate (stable integer spectral dimension with lawful aggregate evolution). Every Φ^+ rule is I^+, but six rules are I^+ without being Φ^+. We conjecture that this asymmetry is structural: for monotone-growth graph dynamical systems with stable eigenspace gaps, Φ-positivity implies I-positivity (Conjecture 3.2), and provide complete computational verification on all 16 rules with zero eigenvalue crossings observed. A perturbation-response experiment confirms the dynamical content of this relationship—perturbed Φ^+ systems return to equilibrium via I^+ trajectories in 108 out of 108 tested cases, with positive dose-response and clean null controls.

1 Introduction

Graph rewrite systems produce diverse dynamical behaviors. Two geometric predicates—the I-predicate, detecting convergent state-space trajectories with compressible dynamics, and the Φ-predicate, detecting stable integer spectral dimension with lawful aggregate evolution—classify these behaviors along independent axes [4]. The predicates are logically independent: neither implies the other in general [4, Theorem 1].

We enumerate all 16 connected DPO graph rewrite rules at signatures $\sigma = 1$ through $\sigma = 4$ and discover a one-sided relationship: every Φ^+ rule is also I^+, but not vice versa. Six rules are I^+ without being Φ^+. The (I^-, Φ^+) cell is empty.

We conjecture this is not a coincidence. For monotone-growth graph dynamical systems with stable eigenspace gaps, Φ-positivity implies I-positivity (Conjecture 3.2). The conjecture is supported by complete computational verification (zero eigenvalue crossings across all rules and seeds) and a partial proof connecting spectral dimension stability to embedding convergence via the Davis–Kahan perturbation theorem [1].

A perturbation-response experiment confirms the dynamical content of this relationship: when Φ^+ systems are perturbed out of geometric equilibrium, they return via I^+ trajectories in 108 out of 108 tested cases, with positive dose-response and clean null controls (0/36). Physical regularity is actively maintained by inference-like dynamics.

Three contributions:

1. **Conjecture 3.2** (computationally verified): Φ-positivity implies I-positivity for monotone growth with stable eigenspace gaps. Verified on all 16 DPO rules with zero exceptions.

2. **Perturbation-response** (Proposition 3.5, computational): Φ^+ equilibrium is maintained by I^+ recovery dynamics (108/108 positive, 0/36 null).

3. **Frequency asymmetry:** $I^+ \supsetneq \Phi^+$ in DPO rule space (15/16 vs. 9/16 non-trivial; 6 I-only rules, 0 Φ-only rules at $\sigma = 4$).

This work was originally motivated by the PRIMO programme (Physical Regularity from Inference in Minimal Ontologies), which conjectured that inference-like behavior precedes physics-like behavior in program space. The present results provide the first computational evidence, with the unexpected finding that the relationship appears structural under growth conditions rather than merely statistical.

2 Preliminaries

We recall both predicates with enough detail to state the theorem precisely. Full definitions, the independence theorem, and threshold analysis are in [4]. The Bayesian forward theorem connecting posterior convergence to the I-predicate is in [5].

2.1 Graph dynamical systems

A *graph dynamical system* is a triple (G_0, R, T) where G_0 is an initial graph, R is a DPO graph rewrite rule, and T is the number of application steps. At each step, R is applied to all non-overlapping matches simultaneously (GPI application). The *trajectory* is $(G_0, G_1, \ldots, G_T)$.

Each rule is evaluated on a family of four canonical initial graphs $G_0 \in \{K_1, K_2, K_3, P_3\}$. A rule is classified as I^+ or Φ^+ only if it satisfies the relevant predicate for at least three of the four initial conditions (3-of-4 majority).

2.2 The I-predicate

Three embeddings map graph trajectories to matrix sequences: e_L (Laplacian eigenvectors), e_R (random projection), and e_D (degree profile). Each embedding e produces a sequence $X_t^e \in \mathbb{R}^{n \times d}$.

Convergence criterion. Compute the cosine-to-final sequence $c_t^e = \cos\theta(\mathrm{col}(X_t^e), \mathrm{col}(X_T^e))$. The Kendall rank correlation $\tau_{\text{to_final}}(e) = \tau(\{c_t^e\}, \{t\})$ measures monotonic approach to the final embedding. The convergence gate requires $\tau_{\text{to_final}} > \tau^* = 0.5$ under at least one embedding.

Compression gate. The trajectory compression ratio $\rho =$ (zlib compressed)/(raw serialized) must satisfy $\rho < \rho^* = 0.85$.

$I^+ =$ compression gate AND convergence under at least one embedding.

2.3 The Φ-predicate

Spectral dimension stability. Compute d_s via log-log regression on the cumulative Laplacian eigenvalue distribution [4]. Require: $\mathrm{std}(d_s) < \sigma_{d_s}^* = 0.08$ over the trajectory, and d_s within 0.5 of an integer.

Lawful evolution. Best polynomial fit to at least one aggregate quantity has residual $< \delta^* = 0.15$.

Curvature homogeneity. Jaccard coefficient of variation of Ollivier–Ricci curvature $< \kappa^* = 1.0$.

Φ^+ = stable spectral dimension AND (lawful evolution OR curvature homogeneity).

2.4 Independence

The predicates are logically independent [4, Theorem 1]. Witnesses: grid_growth (I^+, Φ^+), complete_bipartite (I^+, Φ^-), fixed_grid_noise (I^-, Φ^+), er_random (I^-, Φ^-). The conditional conjecture does not contradict independence—it identifies a structural condition (monotone growth) under which one direction of the implication holds. The fixed_grid_noise witness for (I^-, Φ^+) violates monotone growth.

3 Main results

3.1 Conjecture: Φ-positivity implies I-positivity for monotone growth systems

Conditions. A graph dynamical system (G_0, R, T) is a *monotone growth system* if it satisfies:

(M1) *Monotone growth.* $|V(G_t)|$ and $|E(G_t)|$ are non-decreasing in t.

(M2) *Bounded perturbation per step.* For a DPO rule with right-hand side of size r applied via GPI to a graph G_t with matching of size m:

$$\|\Delta A_t\|_F \leq r \cdot \sqrt{2m}.$$

(M3') *Eventual eigenspace gap stability.* There exists t_1 such that for all $t \geq t_1$, the cluster gap satisfies $\mathrm{gap}_k^{\mathrm{cluster}}(L_t) \geq \gamma > 0$.

Remark 3.1 (On M3'). The cluster gap rather than the raw eigenvalue gap is the correct stability measure. Tree-like growth rules produce graphs with high eigenvalue multiplicity (e.g., multiplicity 57–60 at $\lambda = 1$), giving a raw gap of zero, but a large stable cluster gap (minimum 0.56 in our enumeration). The Davis–Kahan theorem [1] applies to eigenspaces, so cluster gap suffices.

Conjecture 3.2 (Conditional implication)**.** *Let* (G_0, R, T) *satisfy* (M1)*,* (M2)*,* (M3')*. If* (G_0, R, T) *is* Φ^+*, then it is* I^+*.*

Conjecture 3.2 is verified computationally on all 16 enumerated DPO growth rules at $\sigma =$ 1–4, with zero exceptions (Section 4). A proof attempt via the Davis–Kahan perturbation theorem is given in Appendix A; it establishes the correct mechanism but the quantitative bound is loose.

Proof outline.

Part I (Embedding convergence). Φ^+ requires stable spectral dimension, which constrains the bulk Laplacian eigenvalue distribution to converge. Under (M1) and (M2), the per-step

embedding perturbation satisfies $\|X^e_{t+1} - X^e_t\|_F \le C_1 \cdot \|\Delta A_t\|_F$ via embedding continuity [5]. Under (M3'), the Davis–Kahan/Wedin $\sin\theta$ theorem [1, 10] gives:

$$\sin\theta_{\max}(\mathrm{col}(X^e_t), \mathrm{col}(X^e_{t+1})) \le \frac{C_1 \cdot \|\Delta A_t\|_F}{\mathrm{gap}^{\mathrm{cluster}}_k(L_t)}.$$

As the graph grows under (M1), the *relative* perturbation $\|\Delta A_t\|_F / \|A_t\|_F \to 0$. The angular perturbation per step therefore decreases, and c^e_t becomes monotonically increasing for t sufficiently large, giving $\tau_{\text{to_final}} > \tau^*$.

Remark 3.3 (Looseness of the Davis–Kahan bound)*.* Direct measurement shows the mean ratio $\|\Delta A_t\|_F / \mathrm{gap}^{\mathrm{cluster}}_k \approx 11$ across Φ^+ rules, and only 3 of 9 rules show a decreasing ratio. The Davis–Kahan bound is loose—it permits large eigenvector rotations per step, yet zero rotations are observed. A tighter argument is needed to convert Conjecture 3.2 into a theorem.

Part II (Compression gate). By [4, Proposition 3], DPO trajectories with polynomial edge growth have compression ratio $\rho(\tau) = O(\log T / T^{\alpha+1})$, which tends to 0. Since Φ^+ under (M1) implies polynomial edge growth with $\alpha \ge 1$, the compression gate is satisfied for T sufficiently large.

Remark 3.4. The fixed_grid_noise rule (I^-, Φ^+) violates (M1). The Dehn-twist construction—a 10×10 toroidal grid with periodic relabeling—is a verified (I^-, Φ^+) counterexample with $d_s = 2.58$, $\sigma_{d_s} = 0$, and all $\tau_{\text{to_final}} \le 0$.

3.2 Perturbation-response: Φ^+ equilibrium is maintained by I^+ dynamics

Proposition 3.5 (Computational)**.** *Let (G_0, R, T) be a Φ^+ DPO system at equilibrium. Perturbing G_T by randomly rewiring a fraction f of edges produces a graph $G^{(f)}_T$ from which the continued trajectory is I^+.*

Protocol.

1. *Burn-in:* Run $T = 30$ steps to reach Φ^+ equilibrium.
2. *Perturbation:* Remove $f\%$ of edges, add the same number of random edges. Three levels: $f = 5\%$, 15%, 30%.
3. *Recovery:* Continue running the rule for 30 steps from the perturbed graph.
4. *Classification:* Apply the I-predicate to the recovery trajectory.
5. *Controls:* (i) Null recovery—apply random graph dynamics instead of the rule (0/36 I^+). (ii) Unperturbed continuation—no perturbation applied (baseline).

Results.

The binary I^+ classification has perfect separation: 108/108 for rule-driven recovery vs. 0/36 for null recovery. The dose-response in mean τ ($0.73 \to 0.80 \to 0.84$) demonstrates that larger perturbations produce stronger convergence dynamics.

Interpretation. Physical regularity is not merely co-occurring with inference-like dynamics. It is *maintained* by inference-like dynamics. Perturbing the geometry triggers an inference-like recovery process. The dose-response confirms this is causal, not coincidental.

Perturbation	I^+	Elevated-I	Mean $\tau_{\text{to_final}}$
5%	36/36 (100%)	15/36 (42%)	0.731
15%	36/36 (100%)	17/36 (47%)	0.797
30%	36/36 (100%)	16/36 (44%)	0.835
Total	**108/108**	**48/108**	—
Null control	0/36 (0%)	—	—

Table 1: Perturbation-response results across all Φ^+ rules.

3.3 Frequency asymmetry

At signature $\sigma = 4$ (11 connected DPO rules):

Cell	Count	Fraction
(I^+, Φ^+)	5	45%
(I^+, Φ^-)	6	55%
(I^-, Φ^+)	0	0%
(I^-, Φ^-)	0	0%

Table 2: Classification at signature $\sigma = 4$.

Across all 16 rules ($\sigma = 1$ through $\sigma = 4$), excluding the Identity rule (I^-, Φ^-): I^+ 15/15 non-trivial rules; Φ^+ 9/15; I-only (I^+, Φ^-): 6 rules; Φ-only (I^-, Φ^+): 0 rules.

4 Computational verification

4.1 DPO rule enumeration

We enumerate all connected DPO graph rewrite rules at signatures $\sigma = 1$ through $\sigma = 4$. Counts: 1, 1, 3, 11 connected rules. See [3] for the full hierarchy theorem.

Rule	σ	I	Φ	Cell
Identity	$1 \to 1$	$-$	$-$	$I^-\Phi^-$
Vertex Sprouting	$1 \to 2$	$+$	$+$	$I^+\Phi^+$
Edge Sprouting (one-sided)	$2 \to 3$	$+$	$+$	$I^+\Phi^+$
Edge Replacement	$2 \to 3$	$+$	$+$	$I^+\Phi^+$
Triangle Completion	$2 \to 3$	$+$	$+$	$I^+\Phi^+$
Path-4 fresh middle	$3 \to 4$	$+$	$+$	$I^+\Phi^+$
Tri+pendant preserved	$3 \to 4$	$+$	$+$	$I^+\Phi^+$
Tri+pendant shifted	$3 \to 4$	$+$	$+$	$I^+\Phi^+$
Tri+pendant fresh hub	$3 \to 4$	$+$	$+$	$I^+\Phi^+$
Diamond fresh vertex	$3 \to 4$	$+$	$+$	$I^+\Phi^+$
Star-3 replacement	$3 \to 4$	$+$	$-$	$I^+\Phi^-$
Star-3 fresh hub	$3 \to 4$	$+$	$-$	$I^+\Phi^-$
Path-4 partial preserve	$3 \to 4$	$+$	$-$	$I^+\Phi^-$
Diamond minus one	$3 \to 4$	$+$	$-$	$I^+\Phi^-$
Diamond preserved	$3 \to 4$	$+$	$-$	$I^+\Phi^-$
K4 completion	$3 \to 4$	$+$	$-$	$I^+\Phi^-$

Table 3: Classification of all 16 connected DPO rules at $\sigma \leq 4$.

4.2 Eigenvalue gap data (Conjecture 3.2 verification)

We measured Laplacian eigenvalue gaps for all 16 DPO rules across all 4 seeds at $T = 30$ (with $T = 60$ for three borderline rules). Key findings:

- **Zero eigenvalue crossings** across all 16 rules, all 4 seeds, all time steps. DPO monotone growth never causes eigenvalue ordering swaps.
- All 9 Φ^+ rules satisfy (M3'). Three tree-like rules have cluster gap ≥ 0.56.
- Two Φ^- rules also have stable eigenspace gaps but fail Φ^+ on spectral dimension instability ($\sigma_{d_s} > 0.08$). Condition (M3') is not the binding constraint.

The $\|\Delta A_t\|_F/\mathrm{gap}_k^{\mathrm{cluster}}$ ratio—the critical quantity in the Davis–Kahan bound—has mean ≈ 11 across Φ^+ rules, and only 3 of 9 rules show a decreasing ratio over time. The Davis–Kahan bound is loose: it permits large eigenvector rotations, yet none are observed. This is why Conjecture 3.2 remains a conjecture.

4.3 Perturbation-response data

See Table 1. The binary I^+ classification has perfect separation: 108/108 for rule-driven recovery vs. 0/36 for null recovery.

4.4 Temporal I-profiles

Temporal I-profiles ($T = 60$, sliding window $W = 8$) tested whether Φ^+ rules exhibit I^+ transients that decay as Φ-positivity stabilizes. Results: 2 of 9 Φ^+ rules show clear majority transients across all 4 seeds. The weak overall signal is explained by the small-signature regime. The perturbation-response protocol (Proposition 3.5) is the more powerful test.

5 Discussion

5.1 Relationship to the original PRIMO conjecture

The PRIMO programme conjectured two claims:

(a) Ordering: $N_I^{\min} < N_\Phi^{\min}$—inference-like behavior appears at shorter program lengths than physics-like behavior.

(b) Equilibrium: Physics-like behavior is the dynamical equilibrium of inference-like processes.

Claim (a) is absorbed by Conjecture 3.2. For DPO growth rules, $\Phi^+ \Rightarrow I^+$, so $N_I^{\min} \leq N_\Phi^{\min}$ holds trivially. But strict inequality cannot be demonstrated in this setting—the first non-trivial rule (Vertex Sprouting, $\sigma = 2$) is both I^+ and Φ^+. The signature level σ is a faithful proxy for program complexity: $K(\rho) = \Theta(\sigma^2)$ for DPO rules [3].

Claim (b) is supported by the perturbation-response result (Proposition 3.5). Conjecture 3.2 provides the structural basis; the perturbation-response shows this is not just a logical implication but an active dynamical mechanism.

5.2 Relationship to Vanchurin

Vanchurin's programme [6, 7, 9, 8] proposes that physical law emerges as the equilibrium of learning dynamics. The present results provide the first discrete, computational test case. Conjecture 3.2 is the discrete analog of Vanchurin's continuous-time result: stable geometry requires convergent dynamics. The key difference: Vanchurin works in continuous metric spaces with gradient dynamics; we work in discrete graph spaces with DPO rewriting. The fact that the structural relationship survives this translation suggests it is not an artifact of the continuous framework.

5.3 Limitations

1. **Small signatures.** Sixteen rules at $\sigma \leq 4$. The frequency asymmetry needs confirmation at higher signatures.
2. **DPO growth rules only.** All enumerated rules are graph-growing. Non-growth systems can be Φ^+ without being I^+.
3. **Threshold sensitivity.** The Φ-predicate threshold $\sigma^*_{d_s}$ was tightened from 0.18 to 0.08 to achieve I/Φ separation at $\sigma = 4$.
4. **The Davis–Kahan gap.** The per-step bound is loose. Converting Conjecture 3.2 to a theorem requires a tighter argument.
5. **Is the I-predicate detecting growth?** Within DPO, every non-trivial rule is a growth rule. Evidence against: in the 33-rule study [4], non-growth rules like sorting_edges are I^+, and the Bayesian forward theorem [5] shows I-positivity captures convergent dynamics independent of growth.
6. **Dehn-twist counterexample.** A 10×10 toroidal grid with Dehn twist is Φ^+ ($d_s = 2.58$, $\sigma_{d_s} = 0$) and I^- (all $\tau_{\text{to_final}} \leq 0$), confirming that (M1) is essential.

5.4 Future work

- Enumerate $\sigma = 5$ ($4 \to 5$) to test frequency asymmetry at larger scale.
- Multi-rule compositions for testing claim (a) strict inequality.
- Find a tight proof for Conjecture 3.2. Candidates: spectral convergence arguments, averaging arguments, or Lyapunov-function approaches.
- Non-DPO program enumeration (binary lambda calculus or Wolfram-style hypergraph rules [11, 2]).

A Proof outline for Conjecture 3.2

A.1 Lemma (GPI perturbation bound)

Lemma A.1. *For a DPO rule with RHS size r applied via GPI to a graph G_t with matching of size m:*

$$\|\Delta A_t\|_F \leq r \cdot \sqrt{2m}.$$

Proof. Each matched subgraph contributes at most r edge modifications, each a rank-2 symmetric update with Frobenius norm $\leq \sqrt{2}$. Since GPI requires non-overlapping matches, the perturbation matrices have disjoint support. For matrices with disjoint support, $\|\sum_i \delta A_i\|_F^2 = \sum_i \|\delta A_i\|_F^2$. Each match contributes $\|\delta A_i\|_F \leq r\sqrt{2}$. Therefore $\|\Delta A_t\|_F^2 \leq m \cdot 2r^2$, giving $\|\Delta A_t\|_F \leq r\sqrt{2m}$. Since $m \leq |E_t|$, we have $\|\Delta A_t\|_F = O(|E_t|^{1/2})$. □

A.2 The Davis–Kahan step

Step 1 (Embedding perturbation). By E1 (continuity): $\|X_{t+1} - X_t\|_F \leq C_1 \cdot \|\Delta A_t\|_F$.

Step 2 (Subspace perturbation via Wedin). By the Wedin $\sin\theta$ theorem [10]:

$$\sin\theta_{\max}(\mathrm{col}(X_t), \mathrm{col}(X_{t+1})) \leq \frac{C_1 \cdot \|\Delta A_t\|_F}{\mathrm{gap}_k^{\mathrm{cluster}}(L_t)}.$$

Step 3 (Ratio decay under growth). Under (M3'), the denominator is bounded below by $\gamma > 0$. The relative perturbation $\|\Delta A_t\|_F / \|A_t\|_F \to 0$ under growth, so the angular perturbation per step tends to zero.

Step 4 (Summability implies convergence). Since the angular perturbation decreases, the cosine-to-final becomes monotonically increasing for t sufficiently large, giving $\tau_{\text{to_final}} > \tau^*$.

A.3 The $\tau_{\text{to_final}}$ bound

Proposition A.2. *If the angular perturbation per step θ_s is summable ($\sum \theta_s < \infty$), then for T sufficiently large:*

$$\tau_{\text{to_final}}(e) \geq 1 - \frac{2t_2}{T-1} > 0.5$$

where $t_2 = \max(t_1, t_2')$ and t_1 is from (M3'), *t_2' is the onset of the summability regime.*

Proof. For $t \geq t_2$, the cosine-to-final c_t^e is non-decreasing by the Grassmannian triangle inequality. The Kendall τ of a sequence non-decreasing in its last $T - t_2$ terms satisfies $\tau \geq 1 - 2t_2/(T-1)$. Setting $T > 4t_2 + 1$ gives $\tau > 0.5 = \tau^*$. This parallels [5, Corollary 1], replacing the Bayesian source of convergence with spectral stability. □

The summability condition is the gap in the proof: experimental data shows the per-step angular bound from Davis–Kahan does not decrease, so summability is not established by the current argument.

B Computational reproducibility

All computations performed in Python 3.12 with NetworkX, NumPy, SciPy. Random seed fixed at 42. Complete source code: `https://github.com/KarolFilipKowalczyk/PRIMO`. 16 DPO rules, 4 seeds, 3 embeddings, $T = 30$ steps per trajectory ($T = 60$ where noted). Total experimental runtime: < 30 minutes on a single GPU (RTX 3050, 4 GB VRAM).

Acknowledgments

This work is part of the PRIMO research programme. Code and data are available at `https://github.com/KarolFilipKowalczyk/PRIMO`.

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